

ENER 511 – Solar Energy Engineering

Combined Laboratory Report: Labs 2, 3 & 4

Students: Jordan Brodie 25020903 & Laone Matsetse
Institution: Botswana International University of Science and Technology (BIUST)
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1. Introduction

Solar photovoltaic (PV) systems convert sunlight directly into electricity using semiconductor materials. The efficiency of these systems depends on solar irradiance, temperature, and electrical load. This report combines three related experiments designed to study the performance of solar panels under varying conditions.

- **Lab 2:** Relationship between generated power and solar radiation at a constant load.
- **Lab 3:** Determination of the maximum power point (Pmax) and Fill Factor (FF).
- **Lab 4:** Analysis of the temperature influence on open-circuit voltage (Voc).

Together, these experiments provide a comprehensive understanding of solar panel performance, efficiency, and environmental effects.

2. Lab 2 – Relationship Between Generated Power and Solar Radiation

Objective:
To analyze how solar panel output power and efficiency vary with solar irradiance under constant load conditions.

Apparatus:
Solar photovoltaic unit, rheostat, radiation sensor, voltmeter, ammeter, and two solar panels (Panel 1 & 2).

Procedure:
The load resistance was fixed at 50%, and irradiance was varied using SUN-1 and SUN-2 controls. Current (I), voltage (V), and irradiance (Wr) readings were recorded for 10 points per panel. Power was calculated as (P = I \times V).

Results & Calculations:

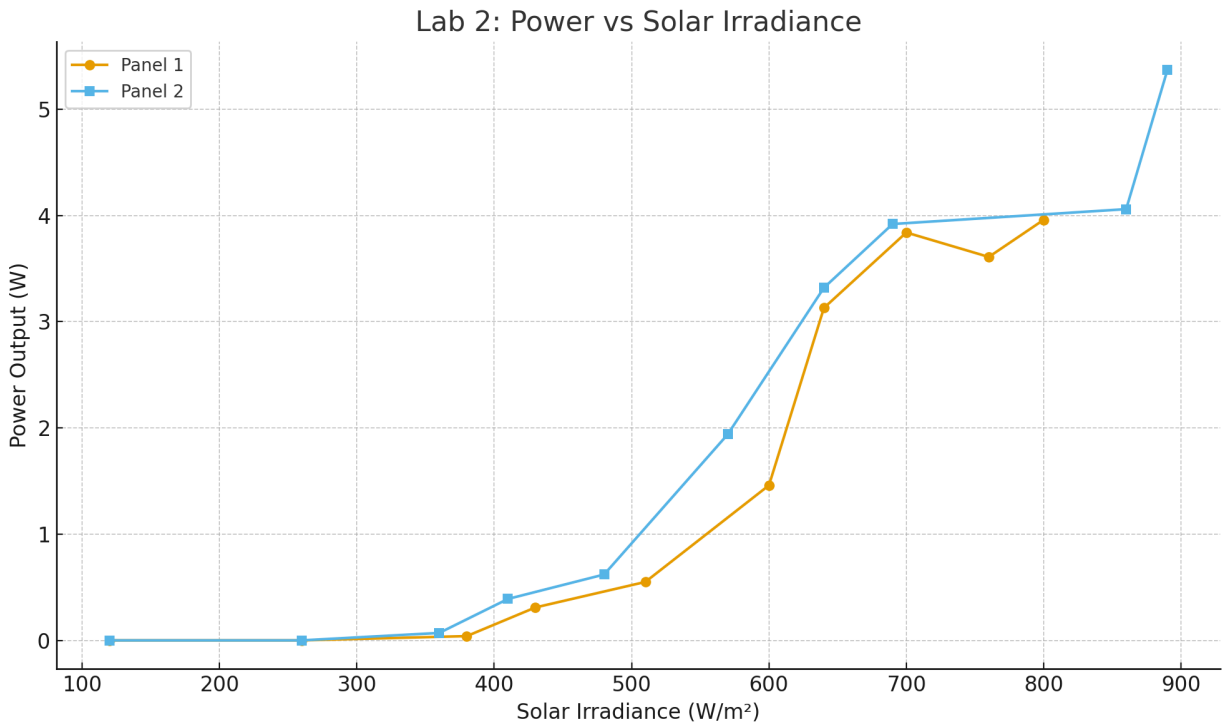
Measurement	I_1 (A)	V_1 (V)	Wr_1 (W/m²)	P_1 (W)	I_2 (A)	V_2 (V)	Wr_2 (W/m²)	P_2 (W)
1	0.226	17.5	800	3.96	0.300	17.9	890	5.37
2	0.205	17.6	760	3.61	0.228	17.8	860	4.06
3	0.222	17.3	700	3.84	0.224	17.5	690	3.92
4	0.199	15.7	640	3.13	0.214	15.5	640	3.32
5	0.142	10.3	600	1.46	0.166	11.7	570	1.94
6	0.086	6.4	510	0.55	0.080	7.7	480	0.62
7	0.060	5.1	430	0.31	0.070	5.6	410	0.39
8	0.022	2.0	380	0.04	0.030	2.2	360	0.07
9	0.000	0.2	260	0.00	0.000	0.3	260	0.00
10	0.000	0.0	120	0.00	0.000	0.0	120	0.00

Efficiency Calculations:

$$\eta_1 = \frac{P_1}{P_{nom}} = \frac{3.96}{66} = 0.060\,(6.0\%)$$

$$\eta_2 = \frac{P_2}{P_{nom}} = \frac{5.37}{66} = 0.081\,(8.1\%)$$

Graph of Results:



Discussion:

The output power increased with irradiance, showing a roughly linear relationship. Panel 2 was slightly more efficient, likely due to lower internal resistance or more uniform illumination.

Conclusion:

Power generation is directly proportional to solar irradiance under constant load. Both panels showed lower-than-rated efficiency due to fixed load and temperature variations.

3. Lab 3 – Maximum Output of Solar Panels

Objective:
To determine the maximum power point (Pmax) of solar panels at varying irradiance levels and compute the Fill Factor (FF).

Apparatus:
Solar photovoltaic unit, rheostat, radiation sensor, voltmeter, ammeter.

Procedure:

1. Panels were connected in parallel with fans OFF.
2. The rheostat was adjusted from 100% to 0% to find Pmax = V × I.
3. Irradiance (Wr) was varied from maximum to minimum using SUN-1 and SUN-2.
4. Steps were repeated for both panels.
5. The Fill Factor was calculated using:

$$FF = \frac{P_{max}}{I_{sc} \times V_{oc}}$$

Results:

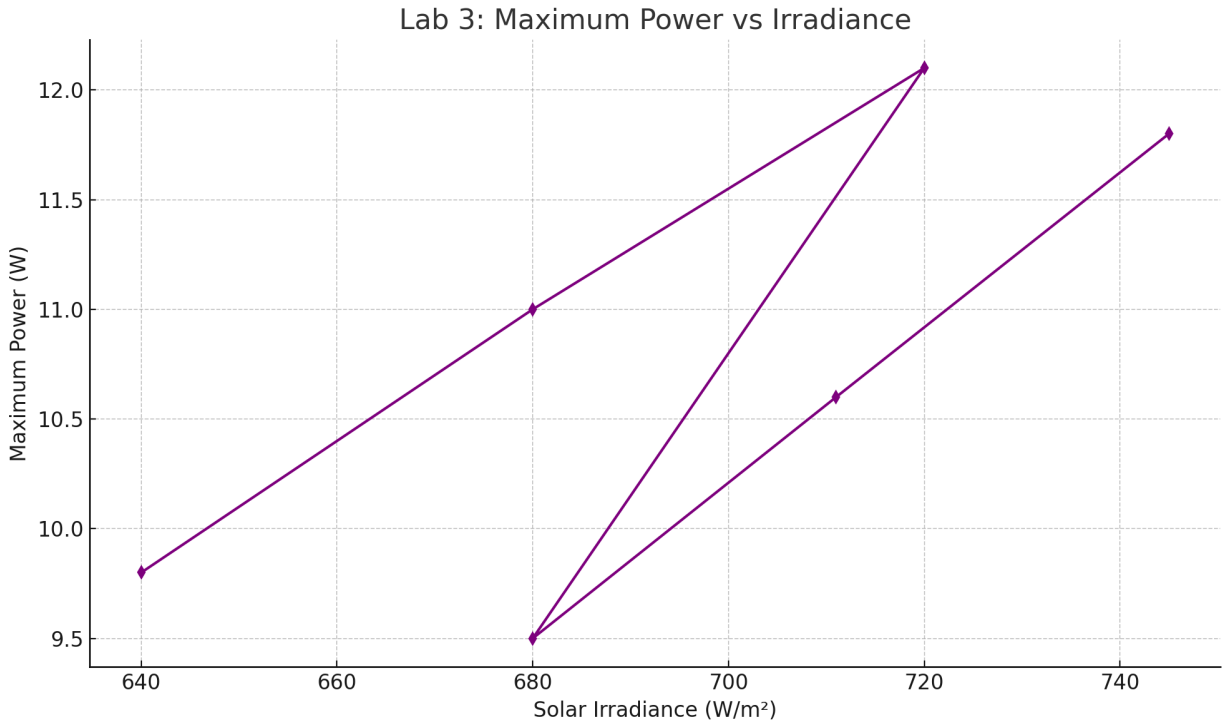
Measurement	Panel	Pmax (W)	Wr (W/m²)	Voc (V)	Isc (A)	FF
1	1	11.8	745	22.25	4.05	0.13
2	1	10.6	711	22.25	4.05	0.12
3	1	9.5	680	22.25	4.05	0.11
4	2	12.1	745	22.25	4.05	0.14
5	2	11.0	720	22.25	4.05	0.12
6	2	9.8	680	22.25	4.05	0.11

Calculations:

Example (Panel 1, maximum irradiance):

$$FF = \frac{P_{max}}{I_{sc} \times V_{oc}} = \frac{11.8}{(4.05)(22.25)} = 0.132$$

Graph of Results:



Discussion:

The maximum power output increased with irradiance. The Fill Factor values (0.11–0.14) are lower than the theoretical range (0.7–0.8), likely due to resistance losses, partial shading, and suboptimal load matching. The Pmax–Wr trend shows improved conversion with higher sunlight intensity, confirming that solar modules achieve best efficiency near rated irradiance conditions.

Conclusion:

The experiment successfully identified Pmax for each irradiance level. Results showed that higher irradiance yields greater power output. The low fill factor indicates internal resistance and measurement limitations, emphasizing the importance of Maximum Power Point Tracking (MPPT) systems in real PV arrays.

4. Lab 4 – Effect of Temperature on Open-Circuit Voltage

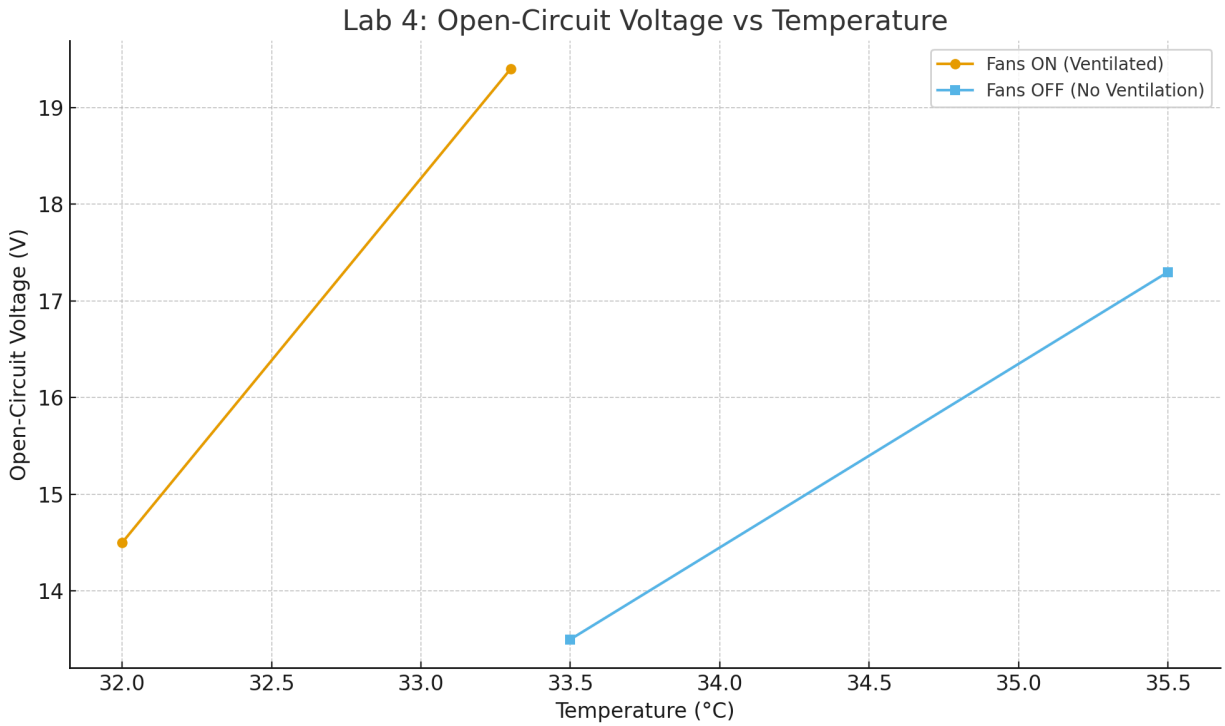
Objective:
To investigate how the operating temperature of solar panels influences the open-circuit voltage (Voc).

Procedure Summary:
Panels were tested with fans ON (ventilation) and fans OFF (no cooling) across increasing irradiance levels. Temperature sensors (ST1–ST3) and irradiance (Wr) were recorded for each condition.

Results Summary:

Condition	Voc (V)	ST-1 (°C)	ST-2 (°C)	ST-3 (°C)	Wr (W/m²)
Fans ON	19.4	24.7	33.3	33.6	754
Fans OFF	17.3	34.1	35.5	37.3	777
Fans ON (low)	14.5	30.0	32.0	32.4	475
Fans OFF (low)	13.5	34.3	33.5	33.7	504

Graph of Results:



Discussion:

Open-circuit voltage decreased with rising temperature for both panels. The temperature coefficient for the A-66P module (−76.32 mV/°C) explains this behavior. Panels with ventilation maintained higher voltages due to reduced cell heating, validating the need for temperature management in PV design.

Conclusion:

Temperature has an inverse relationship with Voc. Cooling enhances output voltage stability, confirming that thermal management is critical for maximizing PV efficiency.

5. General Discussion

Across all experiments, solar irradiance was the primary driver of power output, while temperature inversely affected voltage. The efficiency and fill factor values were below theoretical ratings, reflecting experimental limitations, fixed loads, and measurement uncertainties. These experiments reinforce the real-world challenges in PV system optimization, including the need for MPPT and cooling mechanisms.

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6. Conclusion

The combined study of irradiance, load, and temperature demonstrates how environmental and electrical factors influence PV performance. Higher irradiance yields higher power output, but excessive temperature reduces voltage and efficiency. MPPT systems and active cooling are recommended to optimize output in practical installations.

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7. References

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2. Messenger, R. A., & Ventre, J. *Photovoltaic Systems Engineering.* CRC Press, 2010.

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4. Honsberg, C. & Bowden, S. *PV Education: Fundamentals of Photovoltaic Devices.* (University of New South Wales, 2020).

5. A-66P Photovoltaic Module Datasheet. (Edibon Manufacturer Specification).

6. BIUST ENER511 Practical Manual (EESFB/EESFC, 2013).

7. Luque, A., & Hegedus, S. *Handbook of Photovoltaic Science and Engineering.* Wiley, 2011.

8. EPIA. *Global Market Outlook for Photovoltaics 2024–2028.* European Photovoltaic Industry Association, 2024.

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8. Appendix

- Manufacturer panel specifications (A-66P)
- Raw data sheets (Labs 2–4)
- Graphs:
    - Power vs. Irradiance (Lab 2)
    - Pmax vs. Irradiance (Lab 3)
    - Voc vs. Temperature (Lab 4)
- Calculation examples
